Expectation

The *expectation* $\mathbb{E}[X]$ (also called the mean) of a random variable X is defined as follows, when the relevant integral or sum exists:

Continuous

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

where $f_X(x)$ is the *density* (or p.d.f.) of X, which is zero outside the support of X.

Discrete

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x f_X(x).$$

where $\mathcal{X}$ is the set of possible values of X, and $f_X(x) = P(X = x)$ is the p.m.f. of X.

When it exists and is finite, $\mathbb{E}[X]$ is a real number, not a random variable.

More generally, the expectation of a function $g(X)$ is

Continuous

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx.$$

Discrete

$$\mathbb{E}[g(X)] = \sum_{x \in \mathcal{X}} g(x) f_X(x).$$

Notation: By convention, $\mathbb{E}\, X^2$ means $\mathbb{E}[X^2]$ whereas $(\mathbb{E}\, X)^2$ is the square of the expectation. In general, these are different! Also, there is no difference between brackets [ ] and parentheses ( ) or even braces { }.

Variance and Covariance

The *variance* of a random variable X measures its spread around its mean. It is defined, when finite, as

$$\mathbb{V}\mathrm{ar}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big].$$

The *covariance* of two random variables X and Y is

$$\mathbb{C}\mathrm{ov}(X, Y) = \mathbb{E}\big[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])\big],$$

where the expectation is with respect to their joint distribution.

Some properties are

$$\begin{aligned}
\mathbb{E}[aX + b] &= a\,\mathbb{E}[X] + b, \\
\mathbb{E}[X + Y] &= \mathbb{E}[X] + \mathbb{E}[Y], \\
\mathbb{V}\mathrm{ar}(aX + b) &= a^2\, \mathbb{V}\mathrm{ar}(X), \\
\mathbb{C}\mathrm{ov}(X, X) &= \mathbb{V}\mathrm{ar}(X), \\
\mathbb{C}\mathrm{ov}(aX + b, Y) &= a\, \mathbb{C}\mathrm{ov}(X, Y), \\
\mathbb{C}\mathrm{ov}(X, aY + b) &= a\, \mathbb{C}\mathrm{ov}(X, Y).
\end{aligned}$$

Example (Discrete)**.** Let X be the outcome of a fair six-sided die. Each value $x \in \mathcal{X} = \{1, 2, 3, 4, 5, 6\}$ has probability 1/6, that is $f_X(x) = \frac{1}{6}$ for every $x \in \mathcal{X}$ and 0 else, so

$$\mathbb{E}[X] = \sum_{x \in \mathcal{X}} x f_X(x) = \frac{1}{6} \sum_{x=1}^{6} x = \frac{7}{2} = 3.5.$$

The variance is

$$\mathbb{V}\mathrm{ar}(X) = \frac{1}{6} \sum_{x=1}^{6} \left(x - \frac{7}{2}\right)^2 = \frac{35}{12}.$$

Let $Y = X^2$, then the expectation of Y can be computed in terms of X as

$$\mathbb{E}\,Y = \mathbb{E}\,X^2 = \sum_{x=1}^{6} x^2 f_X(x) = \frac{1+4+9+16+25+36}{6} = \frac{91}{6}$$

If we now want to get the covariance of X and Y, we can use a similar trick ($Y = X^2$ is a function of X)

$$\begin{aligned}\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}\big[(X-\mathbb{E}\,X)(Y-\mathbb{E}\,Y)\big] &= \mathbb{E}\big[(X-\mathbb{E}\,X)(X^2-\mathbb{E}\,X^2)\big] \\ &= \frac{1}{6}\sum_{x=1}^{6}(x-\mathbb{E}\,X)(x^2-\mathbb{E}[X^2]) = \frac{1}{6}\sum_{x=1}^{6}\left(x-\frac{7}{2}\right)\left(x^2-\frac{91}{6}\right) = \frac{245}{12}.\end{aligned}$$

An alternative way to compute the covariance is to re-express it as

$$\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}\big[(X-\mathbb{E}\,X)(X^2-\mathbb{E}\,X^2)\big] = \mathbb{E}\big[X^3\big] - \mathbb{E}\big[X\big]\,\mathbb{E}\big[X^2\big]$$

But now we need the third moment of X, which is

$$\mathbb{E}\,X^3 = \sum_{x=1}^{6} x^3 f_X(x) = \frac{1+8+27+64+125+216}{6} = \frac{147}{2}$$

leading to

$$\mathbb{C}\mathrm{ov}(X,Y) = \mathbb{E}\big[X^3\big] - \mathbb{E}\big[X\big]\,\mathbb{E}\big[X^2\big] = \frac{147}{2} - \frac{7}{2}\frac{91}{6} = \frac{245}{12}.$$

Example (Continuous)**.** Let X be uniformly distributed on $[0,1]$, so its density is $f_X(x) = 1$ for $0 \le x \le 1$ and 0 otherwise. Then

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\,dx = \int_0^1 x\,dx = \frac{1}{2}, \quad \mathbb{V}\mathrm{ar}(X) = \int_{-\infty}^{\infty}(x-\mathbb{E}[X])^2 f_X(x)\,dx = \int_0^1 \left(x-\frac{1}{2}\right)^2 dx = \frac{1}{12}.$$

The mean is the midpoint of the interval, and the variance measures how spread out the values are around that midpoint.

Exercise 5

Problem 1 (Moment Identities)**.** Let X and Y be random variables with finite second moments[1], and let $a, b \in \mathbb{R}$. Use the definitions and the provided identities in this document to show that:

1. $\mathbb{V}\mathrm{ar}(X) = \mathbb{E}[X^2] - (\mathbb{E}\,X)^2$
2. $(\mathbb{E}\,X)^2 \leq \mathbb{E}[X^2]$
3. $\mathbb{V}\mathrm{ar}(aX + bY) = a^2\,\mathbb{V}\mathrm{ar}(X) + 2ab\,\mathbb{C}\mathrm{ov}(X, Y) + b^2\,\mathbb{V}\mathrm{ar}(Y)$

[1]That means $\mathbb{E}\,X^2 < \infty$ and $\mathbb{E}\,Y^2 < \infty$, which ensure that all quantities are well-defined and finite.

Problem 2 (Random Search)**.** Given an array of n numbers, we want to find a position containing the number N, which occures $k \geq 1$ times. The random searh algorithm chooses a position uniformly at random from the n positions and checks its value. If it is not N, the algorithm repeats, choosing again from all n positions. What is the expected number of checks (derived explicitly?)
Note: The same position may be chosen multiple times.

Problem 3 (Normal mean and variance)**.** A continuous random variable X is said to have a *normal distribution* with parameters $\mu \in \mathbb{R}$ and $\sigma > 0$ if its density is given as

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(x-\mu)^2}{2\sigma^2}\Big).$$

Show by direct integration that $\mathbb{E}\,X = \mu$ and $\mathbb{V}\mathrm{ar}(X) = \sigma^2$.
Hint 1: Use the substitution $y = \frac{x-\mu}{\sigma}$ and note that $\int_{-\infty}^{\infty} g(y)\,dy = 0$ if g is odd, that is $g(-y) = -g(y)$ for all $y \in \mathbb{R}$.
Hint 2: The Gaussian integral is $\int_{-\infty}^{\infty} e^{-y^2/2}\,dy = \sqrt{2\pi}$.
Hint 3: Use integration by parts for the variance.

Problem 4 (Carmichael's Die)**.** Denote as X the outcome of a roll of a fair 1729-sided die and let

$$Y = \begin{cases} 1, & \text{if } X \text{ is prime,} \\ 0, & \text{otherwise.} \end{cases}$$

Determin the following (using `R`):

1. the expectation and variance of X,
2. the expectation and variance of Y,
3. and the covariance of X and Y.

Problem 5 (Autoregressive time series)**.** Let Z_i for $i = 1, 2, \ldots$ be iid[2] random variables with mean zero and variance σ^2. Moreover, let X_0 be another random variable independent of the Z_i with mean zero and variance σ_0^2. Now, define recursively

$$X_n = \rho X_{n-1} + Z_n, \qquad n = 1, 2, \ldots$$

for $\rho \in \mathbb{R}$. The time series $X_1, X_2, \ldots$ is called an *autoregressive time series of order one*, AR(1) for short.

1. Show that $\mathbb{E}\,X_n = 0$ for $n = 1, 2, \ldots$.
2. Compute the covariance $\mathbb{C}\mathrm{ov}(X_{n+k}, X_n)$ for $n, k = 0, 1, \ldots$.

[2]Meaning *independently and identically distributed*. In this context, this means that $\mathbb{E}\,Z_i = 0$, $\mathbb{V}\mathrm{ar}(Z_i) = \sigma^2$ for every i and $\mathbb{C}\mathrm{ov}(Z_i, Z_j) = 0$ for $i \neq j$. But note that this holds by assumption for the Z's but *not* the X's!